

Empirical Privacy-Risk Analysis of ECG Representations: Linkability, Attribute Inference, and Waveform Reconstruction

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Abstract—Electrocardiogram (ECG) representations may retain record-specific structure after direct identifiers and raw waveforms have been removed. We treat compact ECG representations as released data products and jointly evaluate, under one matched protocol, utility, same-retained-record segment linkability, linkage-mediated attribute inference, and direct and linkage-conditioned waveform reconstruction. We filtered 12-lead, 10-s PhysioNet ECGs, divided them into overlapping 2-s windows, and extracted 208 handcrafted features. A sample-matched, three-seed comparison evaluated these features, two 8-dimensional linear projections, an 8-dimensional utility-supervised bottleneck, and Gaussian-perturbed bottleneck embeddings. Separate 64-dimensional projection and bottleneck variants assessed dimension sensitivity. Handcrafted features achieved segment- and record-level utility balanced accuracies of 0.832 and 0.860, with pairwise linkability ROC-AUC 0.999. The bottleneck changed these values to 0.865, 0.898, and 0.882; perturbation with noise multiplier $m = 0.194$ and clipping norm $C = 2$ yielded 0.839, 0.894, and 0.807. In a gallery with one true match among 1,000 candidates, Recall@1 fell from 0.887 for handcrafted features to 0.084 for the bottleneck and 0.014 after perturbation. Correctly linked aggregation produced small sex and age effects. A 128-component target PCA itself imposed a minimum attainable PRD of 54.0%; direct segment-level raw-feature MLP decoding, evaluated independently of grouping, reached PRD 96.5%. In dedicated linkage-conditioned runs, raw-feature MLP PRD changed from 96.6% for one anchor to 98.8% with oracle grouping and 99.1% after attacker-selected grouping. Per-segment standardisation and selective feature removal reduced but did not eliminate linkability. Thus, dimensionality reduction and poor waveform reconstruction do not by themselves imply unlinkability. Claims remain bounded to the defined same-record attackers; no cross-session re-identification or end-to-end differential-privacy guarantee is established.

Index Terms—electrocardiography, health data privacy, linkability, attribute inference, waveform reconstruction, representation learning

I. Introduction

Electrocardiogram (ECG) recordings support rhythm analysis, diagnosis, risk stratification, and the development of data-driven clinical tools. Their value for secondary use is accompanied by a privacy concern: physiological morphology, timing, amplitude, and inter-lead structure may preserve stable record-specific information even when direct identifiers are removed. ECG has also

been investigated as a biometric signal, making it important to distinguish clinical utility from the possibility of linkage or recognition [1], [2].

Most privacy evaluations focus on raw signals, model training, or formal mechanisms. A complementary question is what happens when ECG data are shared as derived feature vectors or learned embeddings. For example, a data holder might publish independently generated segment representations without record identifiers so that downstream users can build rhythm classifiers. An attacker who can group segments from the same recording can then aggregate evidence for sensitive attributes. Independently of that grouping success, an attacker can attempt to invert each released segment representation toward its source waveform. A representation can reduce dimensionality without preventing this grouping, or preserve task-relevant information while discarding part of the record-specific structure. This motivates evaluating utility and linkability on the same released object.

Prior ECG privacy work commonly studies biometric identification or evaluates a particular anonymisation mechanism under method-specific release objects and attacker protocols [3]–[7]. These studies establish both the threat and the value of defences, but their heterogeneous designs do not isolate how alternative compact representations change utility and multiple privacy risks under matched conditions. Rather than proposing another anonymisation mechanism, this study treats compact ECG representations as released data products and evaluates them within one controlled, multi-axis protocol.

This paper asks: How well do different ECG representations preserve utility for a record-derived atrial-arrhythmia classification task while reducing linkability, and what additional attribute-inference or reconstruction risks remain? The contributions are:

- 1) a reproducible, sample- and dimension-matched comparison of handcrafted, linear, task-supervised, and perturbed ECG representations on identical retained-record splits;
- 2) a linkability evaluation spanning balanced pairs and a 0.001-prevalence gallery, followed by attacker-selected, oracle, and random aggregation controls for sex and age inference; and
- 3) direct and linkage-conditioned representation-inversion axes with waveform-approximation and random-group controls, together with an explicit

separation between empirical same-record risk, cross-session identity recovery, and formal privacy.

II. Related Work and Problem Formulation

Electrocardiographic signals have long been studied as biometric traits because their morphology can support subject recognition and authentication. Comparative ECG biometric work also shows that performance depends on acquisition and session conditions, making the threat model and evaluation protocol essential to interpretation [1]. Systematic evidence on ECG biometric acquisition similarly reports substantial variation in the number of leads, acquisition configuration, and recording duration [2]. These observations motivate treating ECG privacy as a protocol-dependent empirical property rather than as a single dataset-independent attribute.

Recent work has quantified ECG identification at scale and examined which waveform features support recognition [8], [9]. Transformer-based identification has also been proposed, although that evidence is currently a preprint [10]. Siamese networks have re-identified subjects from anonymised healthcare ECGs using subject-level similarity and motivated-intruder gallery tests [3]; partial knowledge across public datasets enables a related linkage attack [4]. A 2026 preprint additionally evaluated verification and closed- and open-set identification from fiducial feature tables and waveform embeddings [11]; a companion preprint examined cross-dataset, large-gallery, and multi-year performance [12]. These studies target an externally meaningful subject identity or cross-session/cross-dataset linkage. They should not be conflated with the narrower same-retained-record task studied here, but they show why even a bounded grouping result deserves scrutiny.

Privacy-oriented ECG methods span waveform generation, selective perturbation, latent demographic suppression, recoverable representations, and differential privacy. PrivECG uses a generative framework to modify 12-lead waveforms while retaining disease-prediction performance [5]. Lee et al. use attention-derived identification and classification relevance to apply non-uniform noise [6]. The PP-VAE combines adversarial sex, age, and race objectives with clinical-prediction and reconstruction objectives [7], while a morphology-keyed method reports a protected latent representation with self-recoverable decoding on MIT-BIH [13]. REAN, currently a preprint, uses a reconstruction-aware raw-signal anonymiser trained against frozen privacy and utility classifiers [14]. Differential-privacy work on ECG analysis instead emphasises query release, sensitivity, and privacy-budget allocation [15]. Taken together, these studies establish complementary defence strategies, but their different release objects and attacker protocols do not provide a matched comparison of compact representations across balanced pairwise linkage, low-prevalence galleries, staged attribute inference, and direct and linkage-conditioned inversion. This protocol-level gap motivates the controlled evaluation below.

More generally, representation inversion asks how much source information can be recovered from an encoding [16], whereas model inversion can instead exploit access to model outputs [17]. The reconstruction experiment below belongs to the former setting because the attacker receives the released segment representation directly. In ECG reconstruction, PRD is widely used but can be misleading when interpreted alone [18]; comparative quality assessment likewise supports combining complementary measures [19].

The released object in this study is a segment-level representation rather than a raw waveform or database query result. Utility is measured at segment and record levels using a record-derived atrial-arrhythmia label, while the linkability endpoint asks whether two segments belong to the same retained ECG record. The implementation field named `patient_id` is interpreted as `retained_record_id`; the available artefacts do not establish a validated longitudinal patient identity. The resulting claim is deliberately bounded to empirical same-retained-record segment linkability under the stated attacker and gallery protocol. It is not a claim of cross-session patient re-identification resistance, complete anonymisation, or end-to-end differential privacy.

III. Data and Methods

A. Dataset and Cohort Definition

The source manifest contained 45,152 12-lead ECG records from the PhysioNet arrhythmia resource [20]–[23]. The dataset documentation describes one 10-s resting ECG per subject, with rhythm labels assigned and secondarily validated by licensed physicians. One record failed processing, leaving 45,151 retained record identifiers. Each accepted recording had a sampling rate of 500 Hz and a duration of 10 s. Diagnostic codes were mapped to an atrial-arrhythmia or non-atrial group, yielding 9,840 atrial-arrhythmia and 35,311 non-atrial retained records. The implementation assigned the atrial-arrhythmia label when the diagnostic-code field contained either 164889003 (atrial fibrillation) or 164890007 (atrial flutter); records without either code were assigned to the non-atrial group.

B. ECG Preprocessing and Segmentation

Each lead was filtered with a fourth-order Butterworth band-pass filter from 0.5 to 40 Hz. The filtered signal was standardised independently for each lead and complete retained record. The 10-s recordings were divided into 2-s windows with a 1-s step, generating nine segments per record and 406,359 segments in total. The utility label is copied from the source record to its segments and should therefore be interpreted as segment-level prediction of a record-derived label, not as an independently adjudicated diagnosis for every interval.

To test whether complete-record standardisation itself created a linkage shortcut, we repeated feature extraction on all 45,151 retained records after standardising each 2-s segment independently by lead. The complete-record

and per-segment conditions used the same retained-record splits, seeds, feature definitions, utility model, and sampled linkability pairs. This full-dataset ablation isolates shared normalisation context; it does not evaluate a third scheme based on global moments estimated from training records.

C. Handcrafted Feature Representation

For each segment and each lead, 13 features were computed: mean, standard deviation, minimum, maximum, amplitude, energy, skewness, kurtosis, area, absolute mean, root mean square, zero-crossing rate, and number of detected peaks. These generated 156 lead-level features. Four statistics across leads for each feature type added 52 global features, producing a 208-dimensional representation. This same feature space was used for the feature-importance and selective-removal analyses; no additional features or records were introduced for those diagnostics.

D. Representation Transformations

The controlled main comparison included the 208 handcrafted features, principal component analysis (PCA), Gaussian random projection, and a supervised neural bottleneck, with all three compressed representations fixed at eight output dimensions. A prespecified sensitivity analysis repeated the three methods at 64 dimensions. For each seed, one label-balanced subset of 40,000 training segments was reused for every representation; the held-out records, test segments, and sampled linkability pairs were also identical. The encoder used a 128-unit hidden layer before the bottleneck, was trained for at most 25 iterations, and predicted the record-derived atrial-arrhythmia versus non-atrial utility label. All imputers, scalers, projections, encoder parameters, and downstream models were fitted using only the shared training subset. PCA provides a data-dependent linear compression, whereas random projection provides a data-independent geometry-preserving baseline [24]–[26].

E. Utility Task

The utility task classifies each segment as atrial-arrhythmia or non-atrial using the record-derived label. Logistic regression with balanced class weights was trained on each representation after train-fitted median imputation and standardisation. Balanced accuracy is the primary utility metric because the classes are imbalanced. For record-level utility, the positive-class probabilities of the nine test segments from each retained record were averaged and thresholded at 0.5. We report record-level balanced accuracy using the same test records and do not retrain the classifier after aggregation.

F. Threat Model and Claim Scope

The attacker obtains independently released segment representations without record identifiers, learns from

labelled same-record and different-record pairs, and ranks candidates in the same representation space. The pairwise and gallery attacks evaluate this grouping stage; the attribute experiment tests whether attacker-selected grouping improves sex or age inference. A positive relation means membership in the same retained 10-s ECG record, not a validated longitudinal patient identity. Independent recordings from the same person, external identity matching, and cross-session linkage remain untested. The primary reconstruction attacker instead receives one released segment representation directly and does not use pairwise scores, gallery retrieval, or linked aggregates. A complementary linkage-conditioned extension reuses the grouping stage and tests whether an attacker-selected aggregate strengthens reconstruction of the anchor waveform. Direct reconstruction is therefore an independent inversion axis, whereas linkage-conditioned reconstruction is a downstream stage under the same-record grouping protocol. The grouping endpoint remains a study-specific linkability proxy, not a claim of complete anonymisation or formal end-to-end privacy.

G. Pairwise Linkability Attack

The first linkability evaluation asks whether two segments came from the same retained record. Positive pairs contain two segments from the same record, with a minimum separation of four segment positions to reduce trivial overlap. At most three positive pairs were sampled per retained record. Negative pairs contain segments from different records. The attacker receives the element-wise absolute difference between the two representations. An XGBoost classifier with 200 trees, maximum depth 4, learning rate 0.05, row subsampling 0.8, and feature subsampling 0.8 was trained on 2,000 positive and 2,000 negative pairs per split [27]. Unless stated otherwise, negative pairs were sampled uniformly from different retained records. In the hard-negative sensitivity analysis, negative candidates were drawn from the five nearest retained-record neighbours after standardising record-level mean feature profiles.

The primary pairwise metric is ROC-AUC. An ROC-AUC of 0.5 indicates chance ordering, whereas 1.0 indicates complete separation in the sampled pair task.

H. Operational Gallery Attack

Balanced pair testing does not reproduce the low prevalence of true matches in a large candidate set. We therefore constructed an operational-style synthetic gallery. Each query had one positive candidate from the same retained record and 999 negative candidates from different records, giving a positive prevalence of 0.001. One thousand queries were evaluated per seed. We report PR-AUC, Recall@1, and mean reciprocal rank (MRR). The gallery is a controlled stress test rather than a direct simulation of an external patient-identification deployment. For this gallery, random ranking has expected PR-AUC and Recall@1 of 0.001. Because PR-AUC depends on class

prevalence, it is reported alongside the gallery prevalence rather than interpreted as a prevalence-free score [28], [29].

I. Staged Attribute-Inference Attack

Age and sex were parsed from the source headers (45,096 and 45,129 retained records available, respectively). For each of 1,000 test anchors per seed, the attacker ranked three true same-record candidates (segment references 4, 6, and 8 relative to anchor 0) among 99 different-record candidates and selected the top three. Four inputs were compared: one anchor segment (S), the true four-segment aggregate (oracle, O), the attacker-selected aggregate (L), and an anchor aggregated with three random different-record segments (R). Feature vectors were averaged. XGBoost attackers predicted sex, age at least 65 years, or continuous age. Single-segment models were trained on one segment per training record; aggregate models were trained on true same-record aggregates. Two baselines contextualised absolute performance: a no-information predictor used the training majority class or training median age, and a utility-only attacker received the binary record-derived clinical label as its sole input. This separates information inherent to the intended task from additional representation content, and separates the information available under correct grouping from the loss caused by linkage errors.

J. Waveform-Reconstruction Attack

A fixed cohort of 12,000 records was used to test reconstruction independently of the grouping and attribute-inference pipelines. For each retained record, the decoder received the representation of segment reference 0 directly; no pairwise scorer, gallery retrieval, or linked aggregate was used. The target was the corresponding normalised 2-s, 12-lead waveform, downsampled from 500 to 100 Hz. A training-only PCA with 128 components made multi-output decoding tractable. Ridge and two-hidden-layer MLP decoders predicted the PCA coefficients from each released representation; inversion returned the 12-lead waveform. Released inputs were median-imputed and standardised from the training split, and the training PCA scores were standardised before decoder fitting. Ridge used $\alpha = 1$; the MLP used hidden layers 256/128 with early stopping. A training-mean waveform and direct target-PCA reconstruction were negative and approximation controls. The direct target-PCA result is the best reconstruction attainable after the chosen target compression, rather than an attacker. Consistent with the ECG reconstruction literature and its warning against relying on one distortion measure alone [18], [19], we report percent root-mean-square difference (PRD), SNR, Lead II correlation, correlation after a ± 250 -ms lag search, and spectral correlation. Correlation and PRD were also summarised separately for all 12 leads. Per-lead correlation is reported as a mean with its interquartile distribution; because PRD is undefined for a near-zero target lead, such cases were excluded only from the corresponding lead-specific PRD

summary and the valid count was retained. This protocol adapts the general representation-inversion question to the released ECG vectors; its decoder architecture and controls are study-specific. Decoder failure is evidence only against these attacks, not proof of non-reconstructability.

A linkage-conditioned extension used segment 0 as the anchor and segments 4, 6, and 8 as true candidates. For each of 1,000 test anchors per seed, these were mixed with 99 different-record candidates and ranked by the linkage XGBoost model. The inputs were the anchor (S), its true four-segment mean (O), the anchor plus the top three candidates (L), and the anchor plus three random candidates (R). Separate decoders were trained on single anchors and oracle means; all targeted the same anchor waveform with the direct-attack pipeline. Thus, the experiment tests feature-wise grouping, not reconstruction of the full 10-s recording.

K. Gaussian Embedding Perturbation

The eight-dimensional encoder embedding was standardised, clipped to an L_2 radius C , and perturbed with isotropic Gaussian noise:

$$\tilde{z} = \text{clip}_C(z) + \mathcal{N}(0, \sigma^2 I_8), \quad m = \frac{\sigma}{C}, \quad (1)$$

Perturbation strength is parameterised directly by the noise multiplier m , clipping norm C , and standard deviation σ . Both retained operating points use $m = 0.1938$: $C = 2, \sigma = 0.3876$ and $C = 4, \sigma = 0.7752$. These quantities completely specify the empirical release, so no nominal ϵ or δ is assigned. A formal (ϵ, δ) -DP claim would additionally require a defined adjacency relation, justified sensitivity, valid calibration, and composition accounting [30], [31]. The encoder was not trained with DP-SGD, fitted preprocessing parameters were not privately estimated, and repeated releases were not covered by a complete privacy accountant. Consequently, no end-to-end differential-privacy guarantee is claimed.

L. Feature-Importance and Selective-Removal Analyses

On a fixed seed-42 retained-record split, logistic-regression coefficients and XGBoost pair-attacker importances were grouped by feature type and lead. A complementary diagnostic ranked a 150-feature utility-oriented subset with a training-only linear pair classifier, removed the 10 to 50 most linkability-relevant features, and re-evaluated utility and XGBoost linkability. This single-split analysis tests feature overlap rather than final performance.

M. Configuration Selection and Model Capacity

Configurations were treated as fixed measurement-design choices rather than exhaustively optimised deployment models, and their selection was not nested within the final evaluation. An initial single-split development benchmark retained logistic regression as the utility probe because its balanced accuracy exceeded that of XGBoost

(0.8318 versus 0.7754), and retained XGBoost as the pair attacker because its linkability ROC-AUC exceeded that of logistic regression (0.9983 versus 0.9962). A separate two-seed development screen compared 8-, 16-, and 32-D bottlenecks. The 8-D encoder was the smallest candidate, its utility was within 0.0023 of the highest observed value, and its linkability ROC-AUC was the lowest. The Gaussian screen covered noise multipliers 0.0969, 0.1938, and 0.4845. We retained the central multiplier and two clipping norms: $C = 2$ as the high-utility point and $C = 4$ as a stronger-perturbation point at the same multiplier. These points bracket a trade-off; neither is claimed to be optimal. Model families and settings were then frozen for the controlled three-seed comparisons and were not retuned per representation or test seed. This prevents test-seed adaptation but does not remove optimism from same-source, non-nested selection. The 40,000-segment and 25-iteration encoder limits are matched computational budgets, not convergence-based optima; all screens and derived endpoints are versioned with the analysis code.

N. Statistical Reporting and Reproducibility

An 80/20 stratified retained-record holdout prevented segments from a test record entering training. Transformations were fitted on training segments and applied before pair construction. Main results use seeds 42, 123, and 456 and report mean plus or minus standard deviation. Within each seed, every row of the main comparison uses the same label-balanced 40,000-segment training subset, held-out test records, and 2,000 positive plus 2,000 negative linkability pairs. Encoder-based rows additionally share the 25-iteration training budget. Five prespecified paired contrasts subtract the reference metric within each seed: PCA, random projection, and the encoder are compared with raw features, while each Gaussian condition is compared with the unperturbed encoder. We report the mean difference and a two-sided 95% Student- t interval across the three paired seed differences ($n = 3$, two degrees of freedom). With only three seeds, between-run variability and interval width are imprecisely estimated; these are descriptive robustness summaries, not population-level inferential guarantees, and no multiplicity adjustment is made. Attribute effects use 2,000-replicate paired bootstrap intervals within each seed; the reported interval pools paired replicate effects across the three fixed seeds and is conditional on the trained models and splits. The controlled comparison and the operational, attribute, and reconstruction studies use the same seeds and high-level record-disjoint protocol, but their scripts refit models for their respective endpoints. The linkage-conditioned reconstruction extension additionally uses 1,000 dedicated target-cohort queries per seed and fits separate single-view and oracle-group decoders. Results across sections are therefore protocol-matched, not measurements from one identical fitted model instance.

IV. Results

A. Main Utility–Privacy Trade-off

Table I and Fig. 1 report the sample-matched comparison, with all three compressed representations fixed at eight output dimensions. Raw handcrafted features remained almost completely separable. The supervised encoder retained substantially more utility than both linear baselines and was less linkable than PCA. Random projection produced a lower pairwise ROC-AUC than the encoder, but with a large utility loss. Gaussian perturbation reduced linkability further, with an additional segment-level utility cost at the larger clipping norm. Averaging the nine segment probabilities improved record-level balanced accuracy for every representation. In the dimensional sensitivity analysis, increasing the encoder from 8-D to 64-D changed segment BA by -0.0010 while increasing pairwise ROC-AUC by 0.0587.

TABLE I

Controlled main comparison. The three compressed representations are dimension-matched at 8-D. Every row uses the same 40,000 training segments, held-out records, seeds, and pair samples. Record BA averages the nine segment probabilities per retained record. Values are mean plus or minus standard deviation over three seeds. Both Gaussian rows use $m = 0.194$; C denotes the clipping norm.

Representation	Segment BA	Record BA	Pairwise AUC
Raw handcrafted	0.8318 ± 0.0017	0.8597 ± 0.0028	0.9986 ± 0.0002
PCA, 8-D	0.6765 ± 0.0064	0.6860 ± 0.0093	0.9651 ± 0.0029
Random projection, 8-D	0.6436 ± 0.0133	0.6773 ± 0.0172	0.8137 ± 0.0326
Supervised encoder, 8-D	0.8647 ± 0.0027	0.8978 ± 0.0013	0.8825 ± 0.0248
Encoder + Gaussian, $C = 2$	0.8390 ± 0.0071	0.8945 ± 0.0005	0.8074 ± 0.0247
Encoder + Gaussian, $C = 4$	0.8067 ± 0.0115	0.8872 ± 0.0046	0.7233 ± 0.0282

Paired seed contrasts supported these changes. Encoder minus raw was $+0.0329$ [0.0271, 0.0386] in segment BA, $+0.0382$ [0.0284, 0.0479] in record BA, and -0.1162 [-0.1782 , -0.0542] in AUC. Relative to the encoder, the same differences for Gaussian $C = 2$ were -0.0257 [-0.0372 , -0.0142], -0.0034 [-0.0056 , -0.0011], and -0.0751 [-0.0827 , -0.0674]; for $C = 4$, they were -0.0580 [-0.0804 , -0.0356], -0.0106 [-0.0191 , -0.0021], and -0.1591 [-0.2079 , -0.1103], respectively (95% CIs).

Feature-overlap diagnostics showed that linkability importance was concentrated in lead-specific extrema and skewness, whereas utility emphasised RMS and standard deviation. Removing the 50 highest-ranked linkability features reduced ROC-AUC only from 0.9971 to 0.9932 while BA fell from 0.8315 to 0.8023. Hard negatives reduced raw-feature AUC to 0.9927, and per-segment rather than complete-record standardisation reduced it to 0.9897 ± 0.0009 with BA 0.8301 ± 0.0018 . Thus, neither the tested feature removal nor shared normalisation context explained the high linkability.

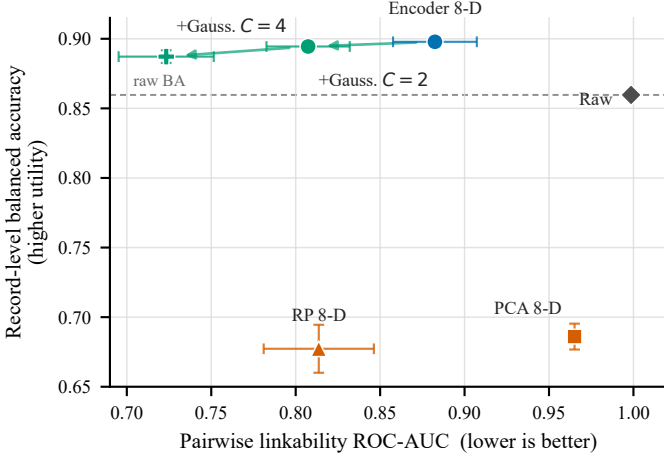


Fig. 1. Sample-matched utility-linkability trade-off (mean plus or minus standard deviation; three seeds). The dashed line marks raw record BA; arrows connect the encoder and Gaussian operating points. Lower AUC and higher BA are preferable. RP denotes random projection.

B. Operational Gallery Comparison

The low-prevalence gallery produced a stricter retrieval view than the balanced pairwise task (Table II). With one positive among 1,000 candidates, the raw representation recovered the true candidate first for 88.7% of queries. Recall@1 fell to 8.4% for the supervised encoder, 1.4% for $C = 2$, and 0.7% for $C = 4$. The corresponding PR-AUC values remained above the 0.001 random-ranking reference, including after perturbation. Pairwise ROC-AUC and Recall@1 are not estimates of the same quantity: the former measures balanced pair ordering, whereas the latter requires recovery of one correct candidate from a low-prevalence gallery. These values come from dedicated protocol-matched reruns and should not be read as the exact fitted instances in Table I.

TABLE II

Operational gallery results with prevalence 0.001. Values are mean plus or minus standard deviation over three dedicated seeds. Random ranking has expected PR-AUC and Recall@1 of 0.001.

Representation	PR-AUC	Recall@1	MRR
Raw handcrafted	0.8285 $\pm$ 0.0036	0.8873 $\pm$ 0.0072	0.9159 $\pm$ 0.0037
Supervised encoder	0.0250 $\pm$ 0.0079	0.0837 $\pm$ 0.0220	0.1513 $\pm$ 0.0263
Encoder + Gaussian, $C = 2$	0.00495 $\pm$ 0.00031	0.0140 $\pm$ 0.0026	0.0473 $\pm$ 0.0010
Encoder + Gaussian, $C = 4$	0.00247 $\pm$ 0.00029	0.0067 $\pm$ 0.0021	0.0278 $\pm$ 0.0066

C. Downstream Attribute Inference

Table III reports absolute attacker performance and two baselines. The no-information classifier had balanced accuracy 0.5, while the training-median age predictor had MAE 15.145 years. A utility-label-only attacker already reached age-group balanced accuracy 0.656 and age MAE 13.825 years, showing that part of the apparent age risk is carried by the intended AF/AFL utility label. With attacker-linked raw features, sex and age-group balanced accuracy reached 0.702 and 0.721, and age MAE fell to 11.162 years. The linked Gaussian representation yielded

0.528, 0.659, and 13.405, respectively, close to the utility-only baseline for age group and with a smaller residual advantage for sex and continuous age. These values measure attribute inference from the released representation under the defined source protocol, not external identity recovery or a probability of individual harm.

TABLE III

Absolute attribute-inference performance (mean plus or minus standard deviation over three seeds). The no-information baseline predicts the training majority class or median age; the utility-only baseline receives only the binary clinical label. “Linked” is the attacker-selected aggregate. Higher BA and lower MAE indicate stronger inference.

Released view	Sex BA	Age ≥ 65 BA	Age MAE (y)
No-information baseline	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	15.145 $\pm$ 0.090
Utility label only	0.515 $\pm$ 0.004	0.656 $\pm$ 0.012	13.825 $\pm$ 0.192
Raw, single	0.686 $\pm$ 0.008	0.707 $\pm$ 0.026	11.563 $\pm$ 0.272
Raw, linked	0.702 $\pm$ 0.019	0.721 $\pm$ 0.017	11.162 $\pm$ 0.160
Encoder, single	0.565 $\pm$ 0.009	0.688 $\pm$ 0.024	12.655 $\pm$ 0.081
Encoder, linked	0.566 $\pm$ 0.010	0.682 $\pm$ 0.026	12.665 $\pm$ 0.196
Gaussian $C = 2$, single	0.536 $\pm$ 0.006	0.665 $\pm$ 0.019	13.303 $\pm$ 0.253
Gaussian $C = 2$, linked	0.528 $\pm$ 0.002	0.659 $\pm$ 0.019	13.405 $\pm$ 0.133

The incremental downstream effects were modest. With raw features, the attacker selected same-record segments with precision 0.9492 ± 0.0040 . Relative to one segment, linked aggregation changed sex balanced accuracy by $+0.0163$ (95% paired-bootstrap CI 0.0036–0.0295) and age MAE by -0.401 years (95% CI -0.532 – -0.264). Although both intervals excluded zero, their magnitudes were small. The age-group balanced-accuracy change was $+0.0134$ (95% CI -0.0003 – 0.0269) and was therefore inconclusive. For the encoder and perturbed encoder, oracle aggregation improved several attribute endpoints, but attacker-selected aggregation did not realise those gains because link-selection precision fell to 0.325 and 0.174. Random different-record aggregation consistently degraded inference, supporting the interpretation that correct linkage, rather than averaging alone, enabled the raw-feature gain.

D. Waveform Reconstruction

The 128-component target PCA retained 0.7167 of training-waveform variance. Its direct reconstruction had PRD $53.97 \pm 0.15\%$ and therefore sets the minimum PRD attainable within the PCA-mediated target pipeline, not a ceiling on error. These are direct segment-level endpoints: they are not conditioned on link selection, and link-selection precision does not enter their calculation. The raw-feature MLP reached PRD $96.49 \pm 0.04\%$, SNR 0.343 ± 0.006 dB, mean lead correlation 0.167 ± 0.001 , and Lead II spectral correlation 0.660 ± 0.003 (Table IV). Ridge PRD was $96.70 \pm 0.03\%$, $99.73 \pm 0.13\%$, and $99.90 \pm 0.03\%$ for raw, encoder, and perturbed-encoder inputs; the corresponding mean-lead correlations were 0.171, 0.052, and 0.045. Encoder and perturbed-encoder decoding approached the training-mean negative control with both model families. Allowing temporal lag increased Lead II correlation to 0.317, 0.256, and 0.237 for the three

MLP inputs and to 0.311, 0.206, and 0.209 for Ridge. These results show weak recovery for the tested PCA-mediated Ridge/MLP attacks; they are not a universal non-reconstructability result.

TABLE IV

Main waveform-reconstruction endpoints for the controls and both decoders (mean plus or minus standard deviation over three seeds). Lower PRD and higher correlation indicate better reconstruction.

Input/decoder	PRD (%)	Mean-lead r	Lead II lag r
Train mean	100.006 $\pm$ 0.002	0.000 $\pm$ 0.001	0.218 $\pm$ 0.004
Target PCA	53.967 $\pm$ 0.147	0.820 $\pm$ 0.002	0.919 $\pm$ 0.001
Raw/Ridge	96.703 $\pm$ 0.027	0.171 $\pm$ 0.002	0.311 $\pm$ 0.003
Raw/MLP	96.493 $\pm$ 0.042	0.167 $\pm$ 0.001	0.317 $\pm$ 0.005
Encoder/Ridge	99.726 $\pm$ 0.135	0.052 $\pm$ 0.003	0.206 $\pm$ 0.007
Encoder/MLP	99.217 $\pm$ 0.313	0.073 $\pm$ 0.008	0.256 $\pm$ 0.009
Gaussian $C = 2$ /Ridge	99.900 $\pm$ 0.030	0.045 $\pm$ 0.003	0.209 $\pm$ 0.005
Gaussian $C = 2$ /MLP	99.734 $\pm$ 0.129	0.050 $\pm$ 0.006	0.237 $\pm$ 0.006

The linkage-conditioned extension used 1,000 target-cohort queries per seed. Mean link-selection precision was 0.952 ± 0.004 for raw features, 0.322 ± 0.033 for the encoder, and 0.174 ± 0.017 after perturbation. MLP PRD (S/O/L/R, in percent) was 96.59/98.77/99.10/100.88 for raw features, 99.22/99.99/99.94/99.98 for the encoder, and 99.76/100.03/100.04/100.05 after perturbation. Raw mean-lead correlation fell from 0.164 (S) to 0.092 (O) and 0.091 (L); the corresponding encoder values were 0.071/0.022/0.023 and perturbed values were 0.049/0.014/0.015. Ridge gave the same qualitative result: grouping did not strengthen anchor inversion.

V. Discussion

Under the evaluated same-retained-record proxy, feature extraction alone did not eliminate linkability: raw features preserved useful label information and almost complete same-record separability. Among dimension-matched 8-D representations, the encoder dominated PCA; random projection reduced linkability further only with a large utility loss. Increasing the encoder to 64 dimensions left utility nearly unchanged but raised AUC from 0.882 to 0.941, so dimension contributed to, but did not explain, the trade-off. Standardisation and selective-removal controls likewise reduced leakage only slightly or at a material utility cost.

In the sample-matched comparison, the supervised bottleneck provided a clearer change because it was learned for the intended utility task. The dedicated operational reruns confirmed that the encoder and Gaussian perturbation substantially reduced candidate retrieval. This is evidence that task-oriented representation design can discard some record-specific information under this protocol, not that supervised embeddings are inherently private. The staged attack produced small downstream changes under successful raw-feature grouping: the paired-bootstrap intervals excluded zero for sex balanced accuracy and continuous-age MAE, but the effect sizes were modest and the binary age-group interval crossed zero. Lower link precision prevented comparable attacker-selected changes for the encoder conditions. Absolute performance nevertheless

remained above both the no-information and utility-only baselines for raw features. Gaussian perturbation approached the utility-only baseline, especially for age-group classification. This distinction separates background attribute predictability from representation-specific exposure and from the incremental effect of attacker-selected linkage.

Direct reconstruction was independent of grouping. Its 128-component target already imposed a substantial PRD floor, and both decoder families recovered little from raw features. Yet raw features remained highly linkable: weak inversion did not imply unlinkability. Neither oracle nor attacker-selected means improved the linkage-conditioned attack; averaging instead removed anchor-specific information. This finding is limited to feature-wise means and does not exclude sequence-aware, generative, morphology-aligned, direct-waveform, or auxiliary-data attacks.

Three limitations especially constrain generalisation. First, configuration screens used the same source and were not nested within the final evaluation; freezing settings prevents per-seed retuning but not selection optimism. Second, three fixed seeds provide coarse between-run uncertainty, so the standard deviations, t intervals, and pooled bootstrap intervals are descriptive and conditional. Third, clinical utility is restricted to binary AF/AFL versus non-atrial classification using a record-derived label; it does not demonstrate preservation of multiclass diagnosis, morphology, prognosis, rare-rhythm detection, or other downstream tasks. In addition, the dataset has one 10-s record per patient, the gallery is synthetic and closed set, and the attribute attack is not external. Reconstruction is limited to one segment position, 100-Hz targets, two decoder families, and feature-wise means. Fixed budgets do not measure scaling, and Gaussian perturbation is not end-to-end differential privacy.

VI. Conclusion

Raw handcrafted ECG features preserved strong same-record linkability and supported attribute inference above both baselines, although attacker-selected linkage added little. A supervised bottleneck outperformed dimension-matched PCA on utility and linkability; Gaussian perturbation further reduced pairwise and gallery linkage while retaining useful performance on the single AF/AFL classification task. The tested direct reconstruction attacks were weak, and oracle or attacker-selected mean grouping did not improve linkage-conditioned inversion. Thus, linkability, attribute inference, and reconstruction require separate evaluation rather than an irreversibility claim. These results remain conditional on non-nested configuration selection, three-seed uncertainty, the narrow utility task, the retained-record protocol, and the tested representation, aggregation, target, and attacker classes; they do not establish cross-session resistance, complete anonymisation, or end-to-end differential privacy.

Ethics Statement

The authors did not collect new data or interact with human participants. The PhysioNet documentation reports that the source studies were approved by the institutional review boards of Shaoxing People's Hospital and Ningbo First Hospital, that informed consent was waived, and that the ECG data were shared publicly after de-identification [20].

Data and Code Availability

The source ECG recordings are publicly available through the PhysioNet resource cited above under the Creative Commons Attribution 4.0 International license. The code, configuration files, and derived result tables used in this study will be made available at <https://github.com/tjpoa/ecg-privacy> before publication, with the final manuscript version archived as a tagged release.

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